

Footings, Retaining Walls and Decks

What carries load into the ground, and what happens when water gets behind something that is holding soil back.

Footing	Transfers the weight of the house to the ground. That is the whole definition — it spreads load over enough soil to carry it.
Frost line	The depth to which the ground freezes. Anything bearing load must sit below it, because freezing soil expands and lifts whatever is above it.

YOU HAVE MISSED THIS

Footings:

Transfer the weight of the house to the ground.

YOU HAVE MISSED THIS

Footings for deck supports:

Should be installed below the frost line.

Retaining walls

A retaining wall holds back soil, and wet soil is far heavier than dry soil. Weep holes at the base let water out through the wall, but they only work if the water can reach them — which means free-draining gravel behind the wall.

YOU HAVE MISSED THIS 2 TIMES

A poured concrete retaining wall with weep holes at the base is rotating. What is the most likely reason?

There is no gravel behind the wall.

The weep holes in the question are a deliberate clue, not a distraction. They tell you drainage was intended. If the wall is still rotating with weeps present, water is being held in the soil and never reaching them, so the gravel is what is missing.

YOU HAVE MISSED THIS

A self-supporting eight-foot concrete block wall leans in half an inch, is not cracked, and is damp with moderate efflorescence. Who is the best specialist to recommend?

A landscape contractor.

Nothing is cracked and nothing is carrying the house — it is a site wall with a drainage problem, and drainage around a site wall is landscaping work. Reach for a structural engineer when the structure of the building is involved, not when a garden wall is damp.

Decks

YOU HAVE MISSED THIS 3 TIMES

What is the function of diagonal bracing on a deck?

It provides additional rigidity to the deck.

Diagonal bracing resists racking — the sideways parallelogram movement a square frame makes when pushed. It adds stiffness, not load capacity. Posts and beams carry the weight; the bracing stops the whole thing swaying.

Say it back

Cover the page. Answer each one out loud.

1. What does a footing do, in one sentence?
2. Where must a deck footing sit, and why?
3. A retaining wall with weep holes is rotating. Most likely cause?
4. Why are the weep holes a clue rather than a red herring?
5. A damp, leaning, uncracked garden wall. Which specialist, and why not an engineer?
6. What does diagonal bracing on a deck actually do?